

Tenant Thermal Comfort - Lessons Learned from Setpoint Optimisation

Tanjong Quay Tower · Office · Singapore

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1. What happened

In March 2024, to accelerate progress towards the building's energy target, space temperature setpoints at Tanjong Quay Tower were raised from 23°C to 25°C across all floors in a single overnight change. Within two weeks, hot complaints rose from about 12 to 96 per month. Two anchor tenants escalated to the asset manager, and several tenants brought in desk fans.

2. Root causes

- No tenant communication: occupants experienced the change without warning or explanation.
- One setpoint for every zone: west-facing perimeter zones with high afternoon solar gain overheated, while interior zones were comfortable.
- Humidity was ignored: relative humidity of 65-70% made 25°C feel considerably warmer.
- VAV box minimum airflows were set too low, so air movement at desks was poor.

3. Recovery plan

- Setpoints were returned to 24°C, then raised in 0.5°C steps every two weeks, zone by zone, only where comfort feedback stayed stable.
- Elevated air speed: ceiling and desk fans providing 0.3-0.5 m/s of air movement allow a higher setpoint at the same perceived comfort.
- Humidity first: AHUs were re-tuned to control the space dew point, keeping relative humidity below 60%.
- Tenant engagement: each tenant nominated a 'comfort champion' and receives a monthly energy and comfort report; green-lease clauses were introduced at renewal.
- Comfort feedback loop: QR codes in each zone let occupants report 'too warm' or 'too cold'; each report creates a BMS work order and reports are reviewed weekly.

4. Results after six months

KPI	Before change	Peak of complaints	After recovery
Hot complaints per month	12	96	31
Average space setpoint	23.0°C	25.0°C	24.5°C
Cooling energy vs 2023	-	-15%	-11%
Tenant satisfaction (survey)	84%	61%	81%

5. Lessons learned

- Never change setpoints building-wide overnight. Small, communicated steps are accepted; surprises are not.
- Humidity matters as much as temperature in a tropical climate.
- Measure comfort, not just energy. A saving that drives tenants away is not a saving.
- People accept changes they understand: explain the why, share the results and give occupants a voice.
- These lessons apply wherever occupants are sensitive to indoor conditions or have contractual comfort expectations.